

Project 3: Network Traffic Analysis

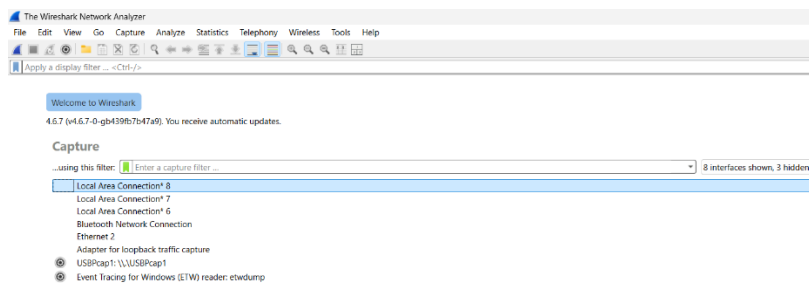
Task 1: Identify Network Information

- Run ipconfig (Windows) or ip addr (Linux).
- Record:
 - **IP Address** (your machine's identifier)
 - **Default Gateway** (router)
 - **DNS Server** (used for domain resolution)

 Take a screenshot of your terminal output.

Task 2: Capture Network Traffic

- Open **Wireshark**.
- Select your active interface (Wi-Fi/Ethernet).
- Start capturing packets.



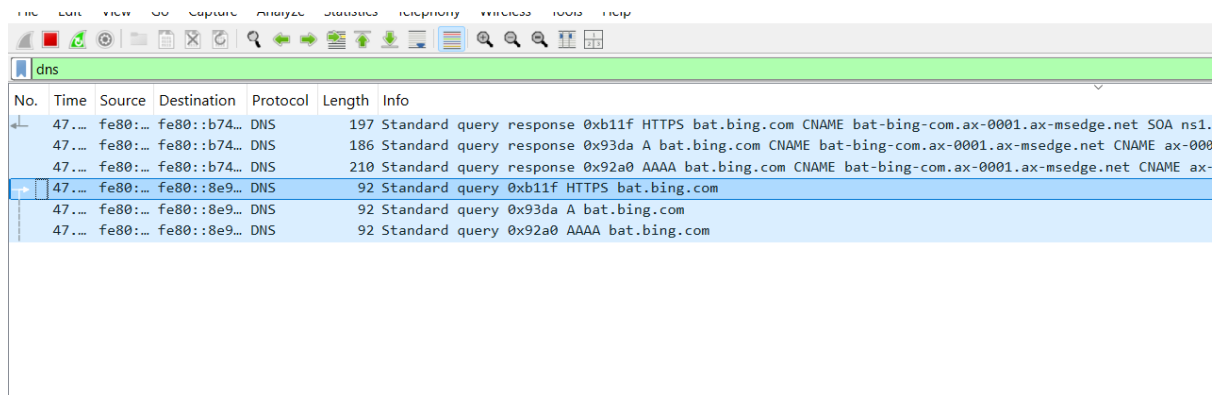
Task 3: Visit TestFire



- Go to <https://testfire.net>.
- Open the **Login page**.
- Stop the capture.

Task 4: Analyse DNS

- Apply filter: dns.
- Find query for testfire.net.
- Note re



No.	Time	Source	Destination	Protocol	Length	Info
47...	...	fe80::b74...	fe80::b74...	DNS	197	Standard query response 0xb11f HTTPS bat.bing.com CNAME bat-bing-com.ax-0001.ax-msedge.net SOA ns1.
47...	...	fe80::b74...	fe80::b74...	DNS	186	Standard query response 0x93da A bat.bing.com CNAME bat-bing-com.ax-0001.ax-msedge.net CNAME ax-000
47...	...	fe80::b74...	fe80::b74...	DNS	210	Standard query response 0x92a0 AAAA bat.bing.com CNAME bat-bing-com.ax-0001.ax-msedge.net CNAME ax-
47...	...	fe80::8e9...	fe80::8e9...	DNS	92	Standard query 0xb11f HTTPS bat.bing.com
47...	...	fe80::8e9...	fe80::8e9...	DNS	92	Standard query 0x93da A bat.bing.com
47...	...	fe80::8e9...	fe80::8e9...	DNS	92	Standard query 0x92a0 AAAA bat.bing.com

solved IP address.

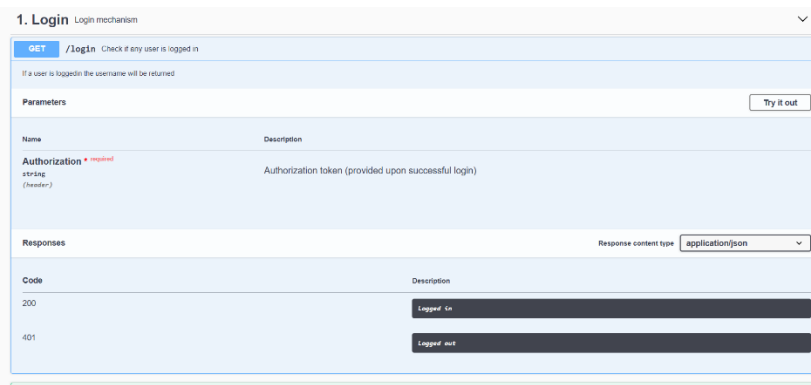
Task 5: Analyse TCP

- Apply filter: tcp.
- Locate the **three-way handshake**:
 - SYN → client initiates
 - SYN-ACK → server responds
 - ACK → client acknowledges

 Screenshot showing SYN, SYN-ACK, ACK packets.

Task 6: Analyse HTTP/HTTPS

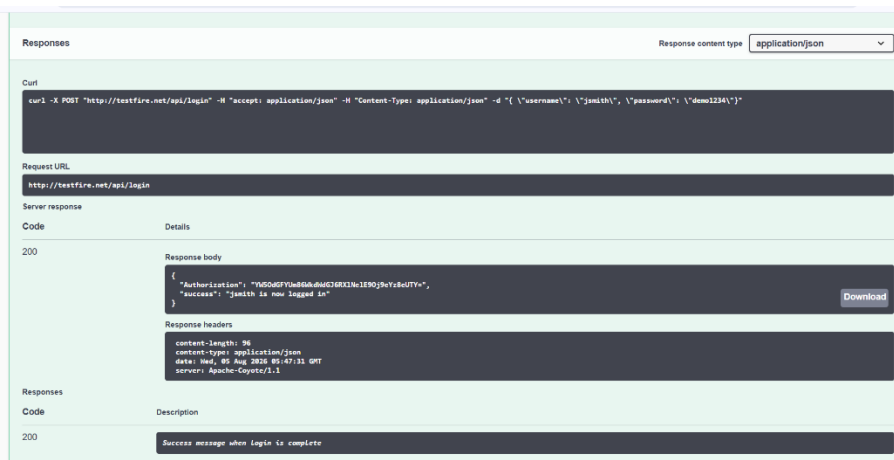
- Open **Browser Developer Tools** (F12 → Network).
- Reload login page.
- Record:
 - Request URL
 - Request Method (GET/POST)
 - Response Status Code (200, 302, etc.)
 - Request Headers
 - Response Headers



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Task 7: Analyse Cookies

- In Developer Tools → Storage/Network tab.
- Identify cookies (e.g., JSESSIONID).
- Explain purpose:
 - Session management
 - Authentication state
 - User tracking/preferences



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Task 8: Report

Structure your report like this:

- 1. Introduction** Objective of the analysis.
- 2. Methodology** Tools used (Wireshark, Chrome/Firefox).
- 3. Findings**
 - IP addressing info

- DNS resolution
- TCP handshake
- HTTP/HTTPS headers
- Cookies

4. Screenshots & Observations Annotated captures for each task.

5. Conclusion How DNS, TCP, HTTP/HTTPS, and cookies work together to enable secure communication and login functionality.